

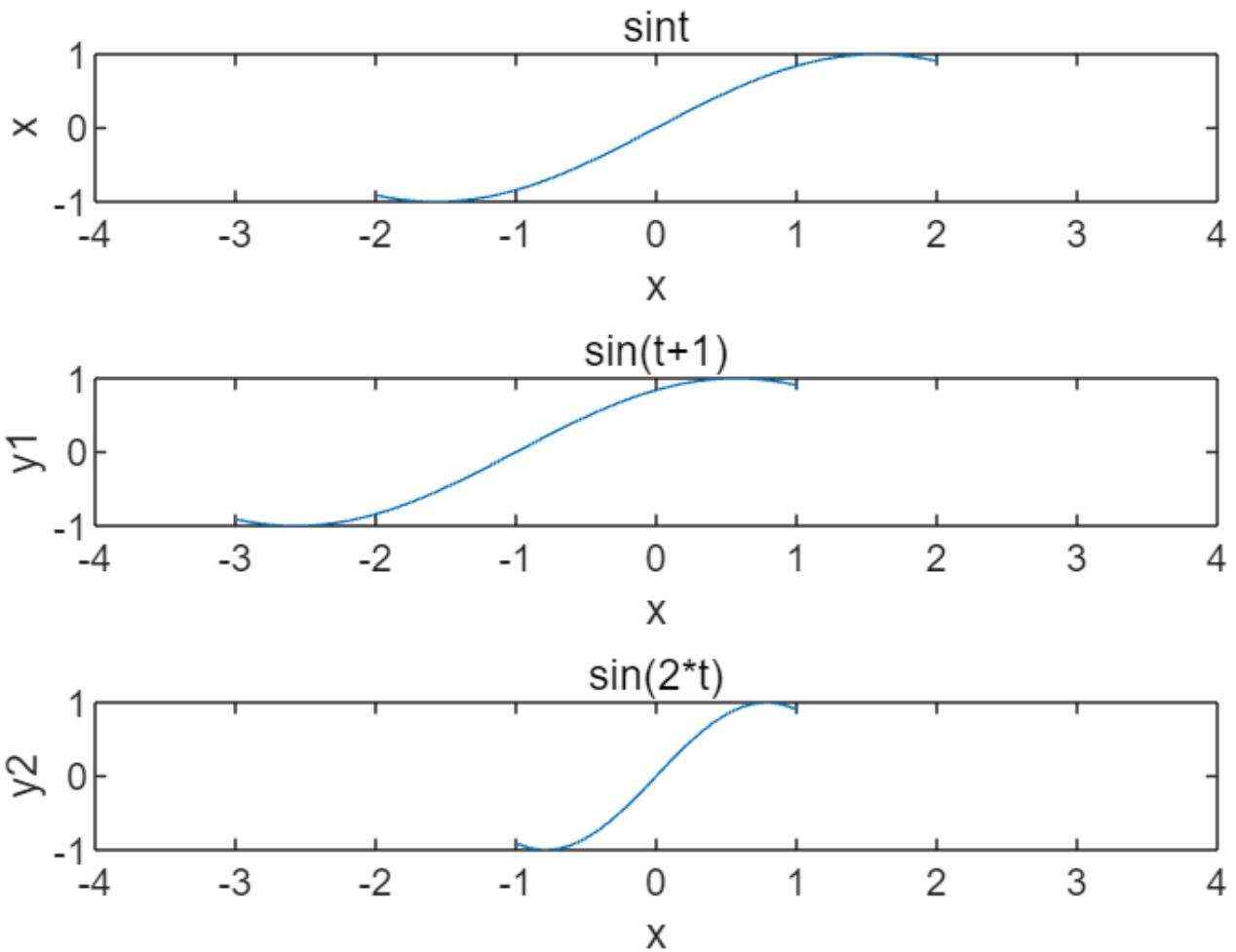
1 : An CT signal is,

$$x(t) = \begin{cases} \sin(t), & t = -2:0.1:2 \\ 0, & \text{otherwise} \end{cases}$$

and : $y_1(t)=x(t+1)$, $y_2(t)=x(2t)$

Please plot $x(t)$, $y_1(t)$ and $y_2(t)$. Requirements: Use subplot to draw three subgraphs, with x axis, y axis labels and titles, and the x axis range of the three subgraphs is the same [-4 4]---Tip, use xlim.

```
clear
t = -2:0.1:2;
t1 = -3:0.1:1;
t2 = -1:0.1:1;
figure
subplot(3,1,1)
plot(t,sin(t))
xlim([-4,4])
xlabel('x'),ylabel('x'), title('sint')
subplot(3,1,2)
plot(t1,sin(t1+1))
xlim([-4,4])
xlabel('x'),ylabel('y1'), title('sin(t+1)')
subplot(3,1,3)
plot(t2,sin(2*t2))
xlim([-4,4])
xlabel('x'),ylabel('y2'), title('sin(2*t)')
```



2 : The unit impulse response of an LTI system is known as follows:

$$\begin{cases} h[n] = 1, & 0 \leq n \leq 5 \\ 0, & \text{else } n \end{cases};$$

1) The difference equation of the system is $y = x[n] + x[n-1] + x[n-2] + x[n-3] + x[n-4] + x[n-5]$.

Give its a and b coefficients :

```
clear
a=[1];
b=[1,1,1,1,1,1];
```

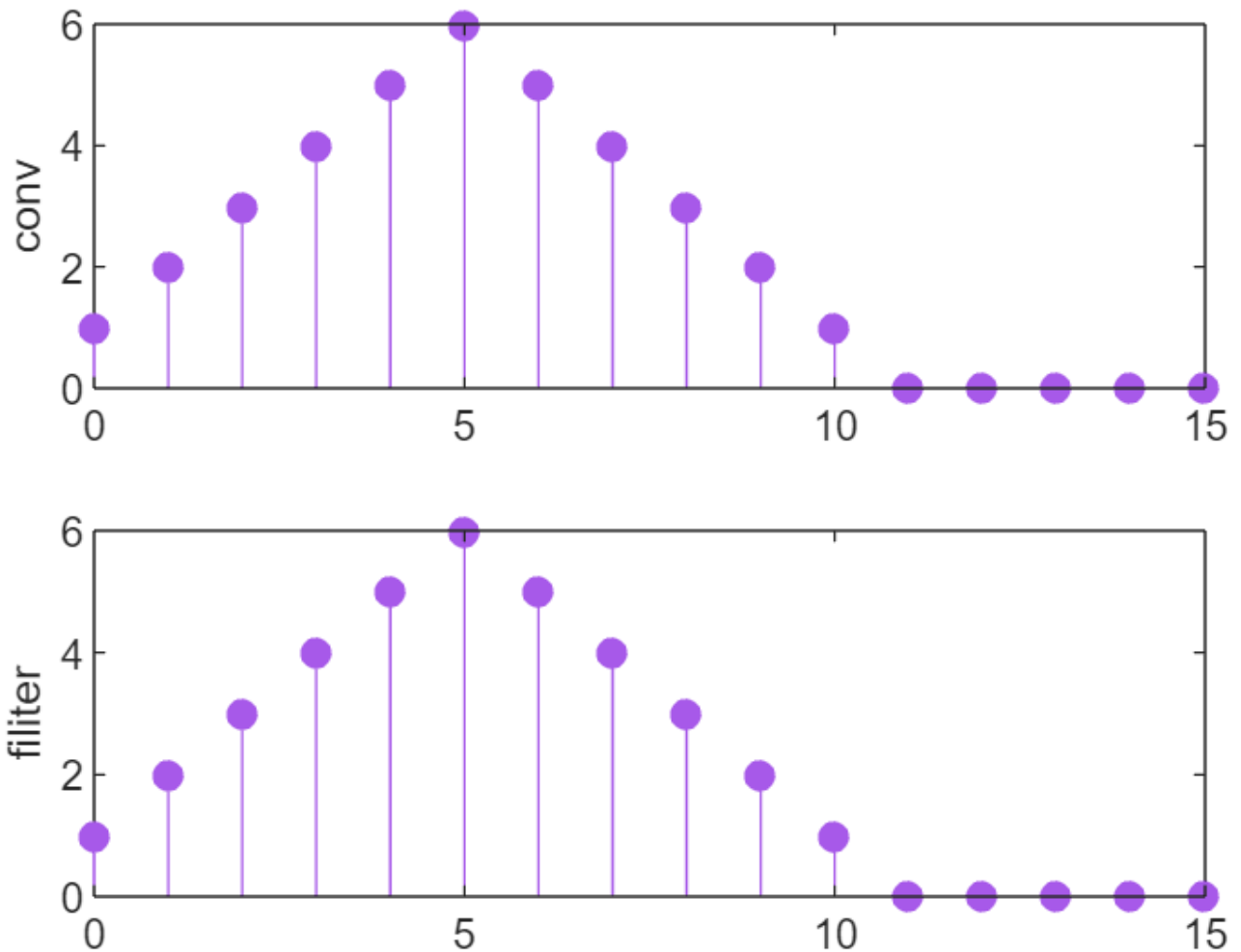
2) When the system's input is : $\begin{cases} x[n] = 1, & 0 \leq n \leq 5 \\ 0, & \text{else } n \end{cases}$, find the output of the system and draw its picture. Please solve the problem in two ways, and draw the two results in one picture with two subgraphs, x axis range [0 15];

figure

```

xn = [1,1,1,1,1,1,0,0,0,0,0,0,0,0,0,0];
lx = 0:1:15;
hn= [1,1,1,1,1,1,0,0,0,0,0,0,0,0,0,0];
lh = 0:1:15;
ly = lx(1)+lh(1):lx(end)+lh(end);
y1 = conv(xn,hn);
subplot(2,1,1)
stem(ly,y1, 'filled', 'Color',[167,89,233]/255);
xlim([0,15])
ylabel('conv')
subplot(2,1,2)
y2 = filter(b,a,xn);
stem(lx,y2, 'filled', 'Color',[167,89,233]/255);
xlim([0,15]);
ylabel('filiter');

```



3 : Comprehensive application questions

1) There is a periodic input signal x , whose period $N=20$ and its DTFS coefficient is:

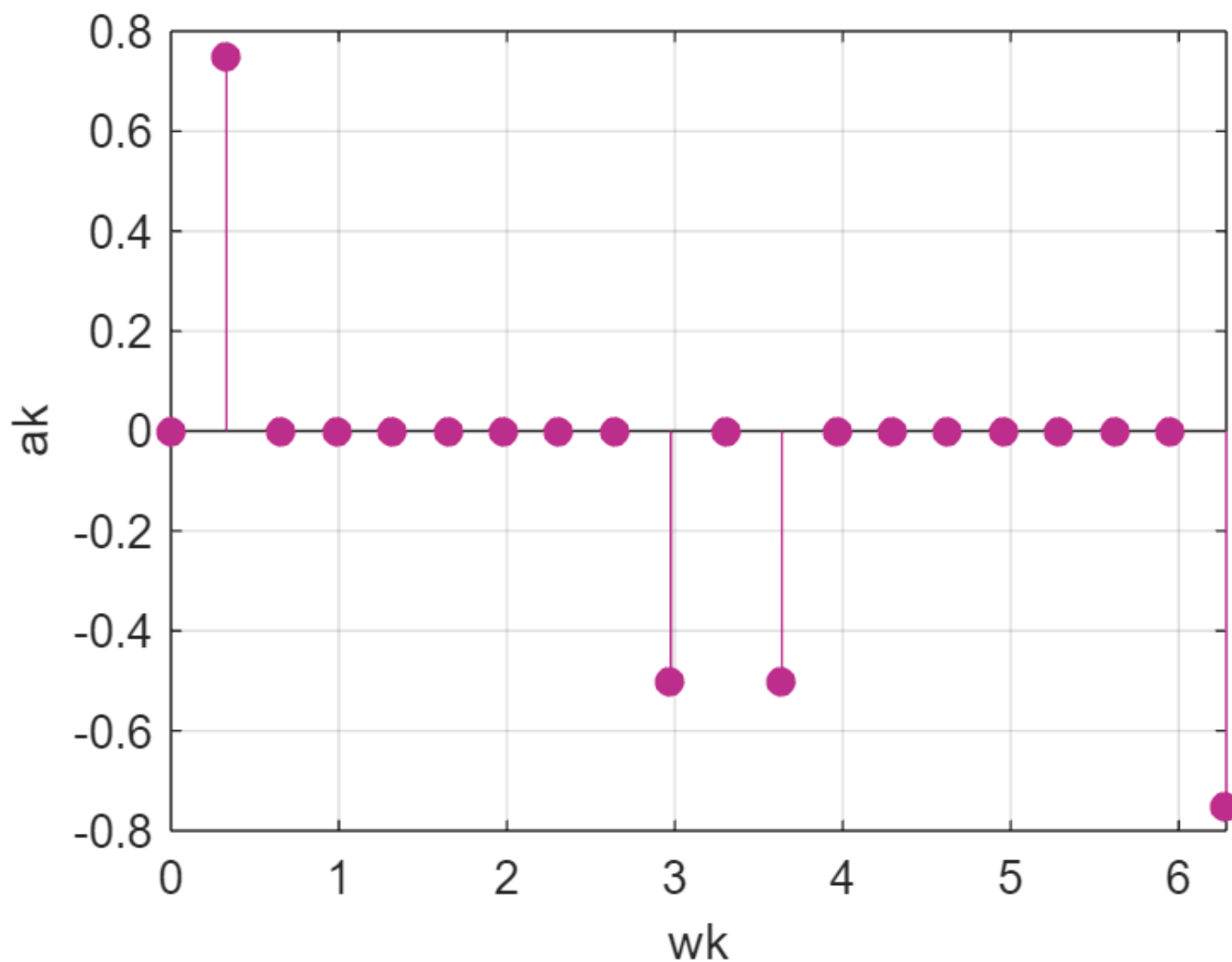
$$a_k = \begin{cases} 3/4, & k = \pm 1, \\ -1/2, & k = \pm 9, \\ 0, & \text{otherwise.} \end{cases}$$

Please draw the spectrum coefficient diagram of the input signal x with w_k as its x axis;

Hints :

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 n} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/N)n}$$
$$a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n}$$

```
clear
ak=[0 0.75 0 0 0 0 0 0 0 -0.5 0 -0.5 0 0 0 0 0 0 0 -0.75];
wk=linspace(0,2*pi,20);
figure
stem(wk,ak,'filled','Color',[189,45,139]/255);
xlabel('wk');
ylabel('ak');
grid on;
```

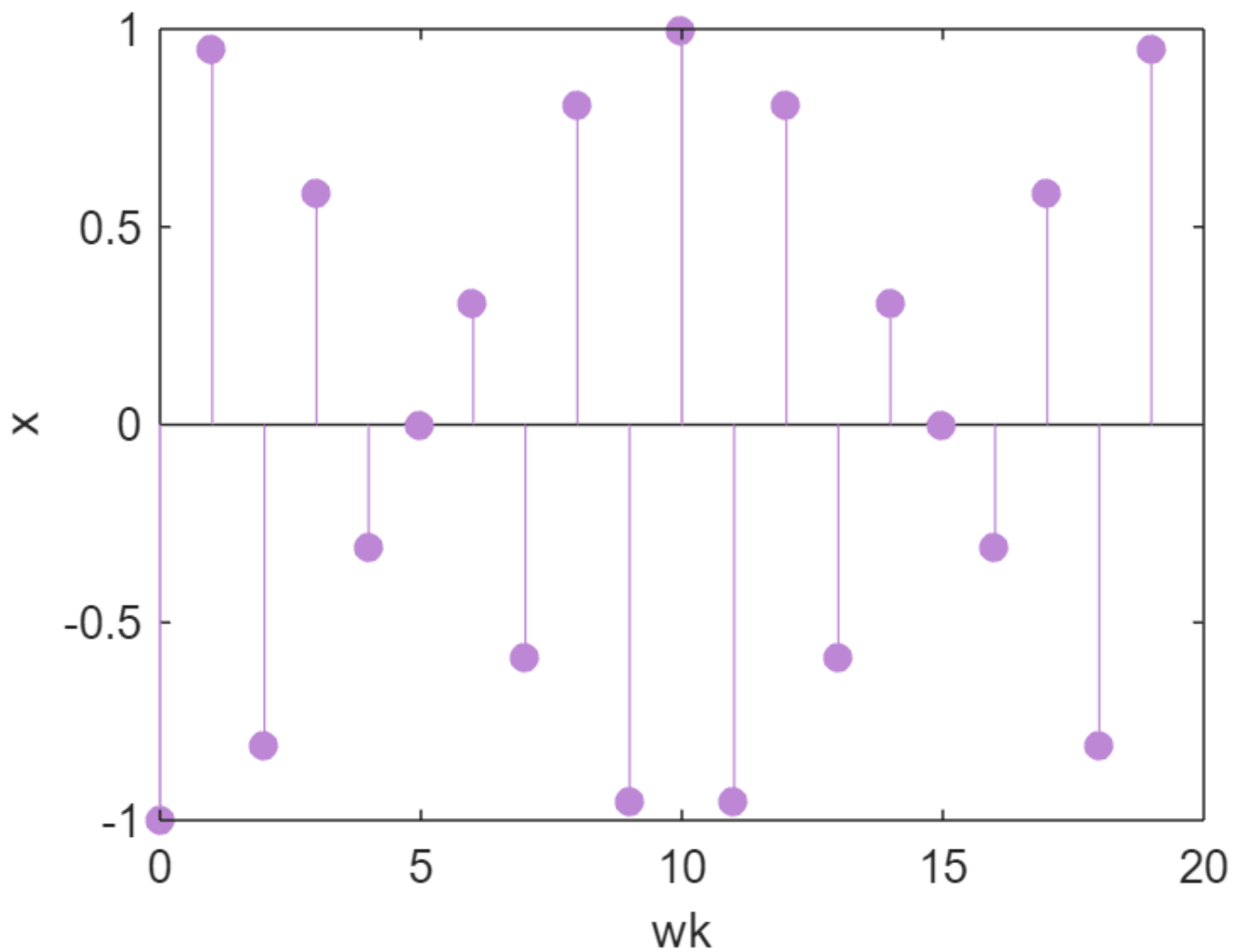


2) Synthesize a period of signal x and draw its picture with x axis range is $0 \leq n \leq 19$.

```
figure
x3=ifft(ak)*20;
nx3=0:19;
stem(0:19,x3,'filled','Color',[189,135,213]/255);
```

警告：仅使用复数数据的实部。

```
xlabel('wk')
ylabel('x')
```



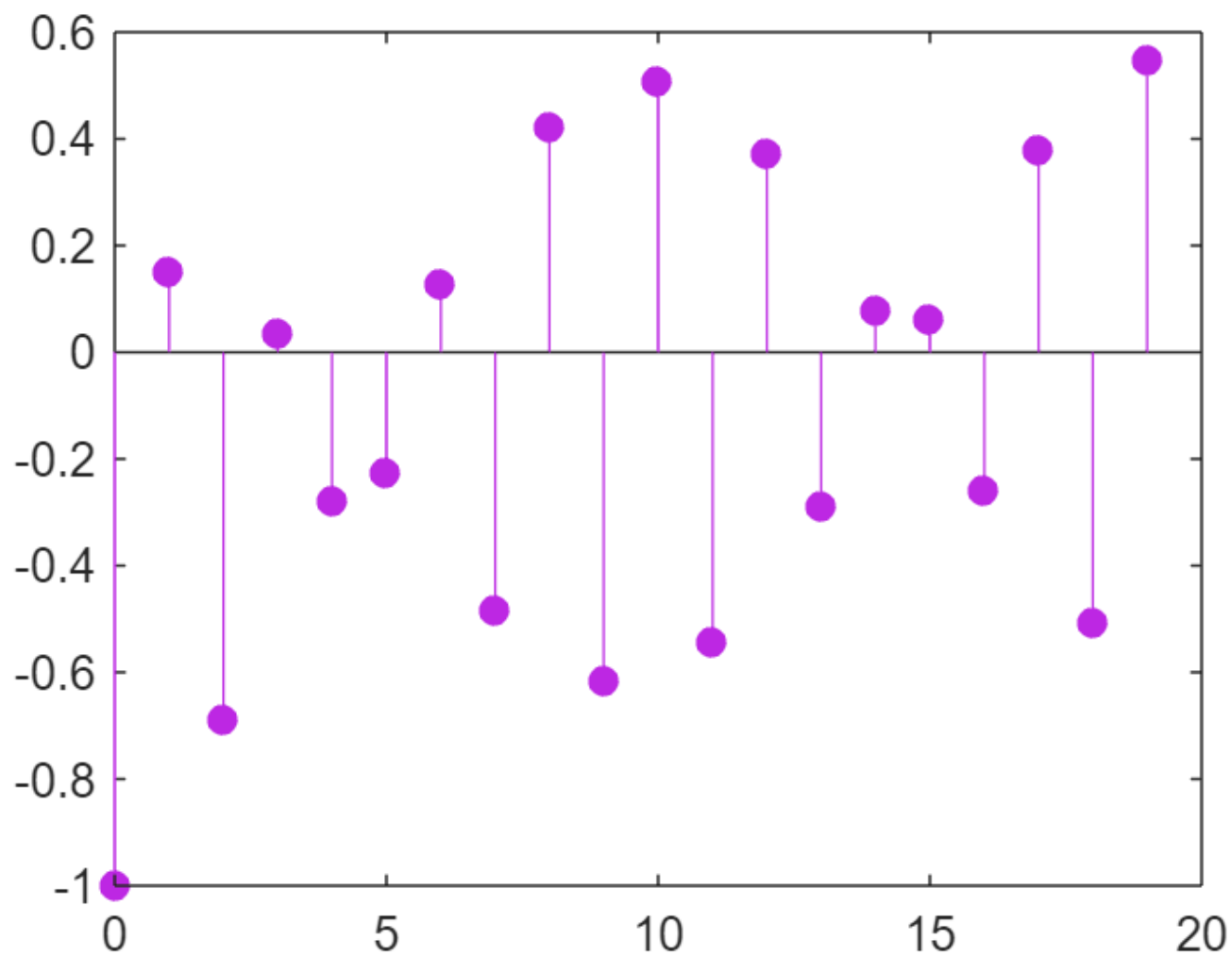
3) We have a DT LTI System as follows:

$$y[n] - 0.8y[n-1] = x[n],$$

Please calculate the output y_1 of the system at the input x directly in the time domain, and draw it.

```
a = [1, -0.8];
b = [1];
y1 = filter(b, a, x3);
figure
stem(nx3, y1, 'filled', 'Color', [189, 39, 227]/255);
```

警告：仅使用复数数据的实部。



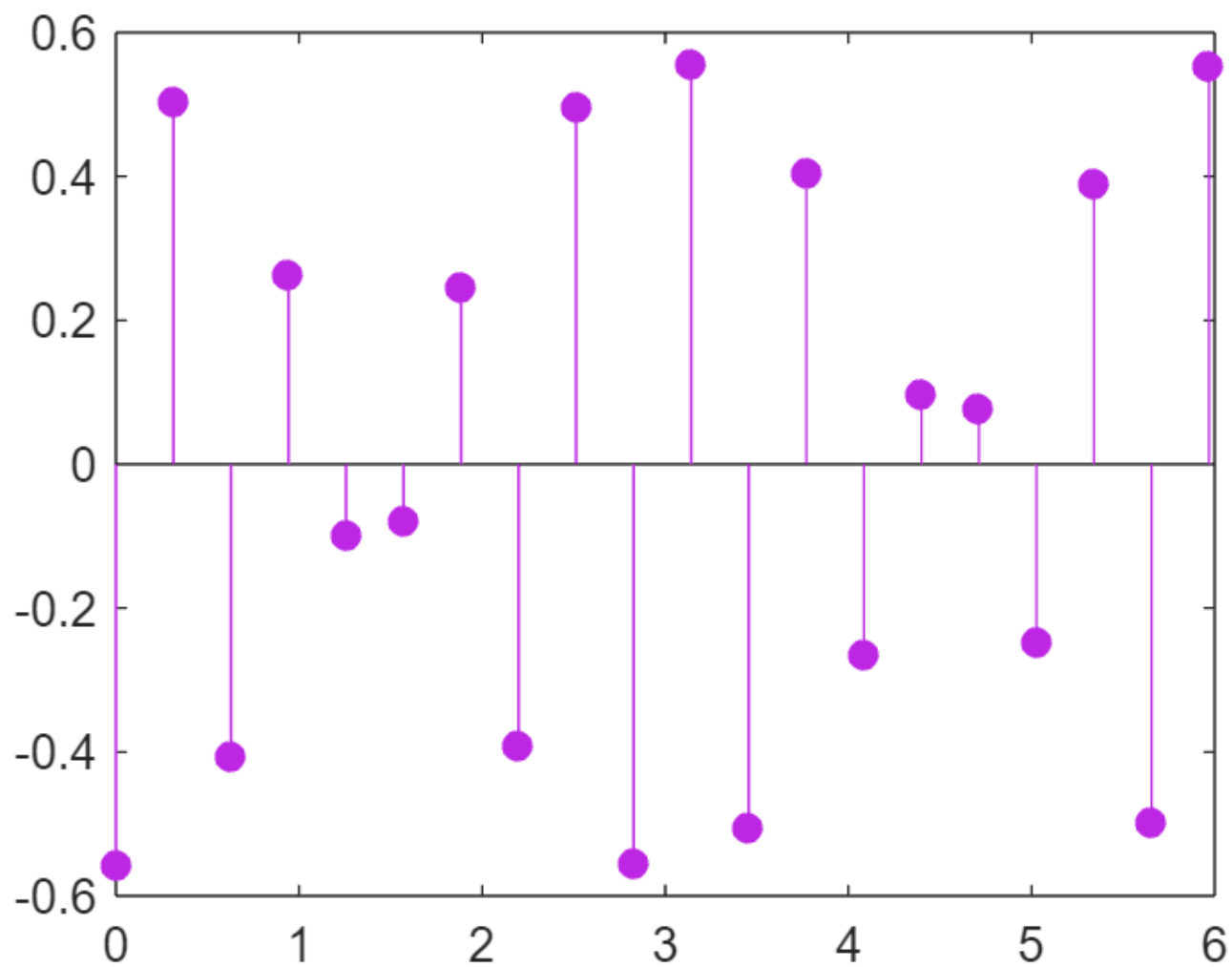
5) Please calculate the output y_2 of the system at the input x in the frequency domain, and draw it.

$$x[n] = \sum_k a_k e^{j\omega_k n} \longrightarrow \text{DT LTI Systems } (h[n]) \longrightarrow y[n] = \sum_k a_k H(e^{j\omega_k}) e^{j\omega_k n}$$

Hints:

```
a = [1, -0.8];
b = [1];
[H omega] = freqz(b, a, 20, 'whole');
Y2=H.'.*ak;
y2 = ifft(Y2)*20;
figure
stem(omega,y2, 'filled', 'Color', [189,39,227]/255);
```

警告：仅使用复数数据的实部。



6) Compare y_1 with y_2 . Are they the same? Why?

不一样。因为我们直接对信号进行时域计算，算出的是准确值，但是由于使用 `freqz` 函数设置了 $n=20$ ，即取前 20 个频率响应点数，并未取全，因此 y_1, y_2 不一样。如果将 n 值设置得更大， y_2 与 y_1 就会越接近。